

# TEJAS BILLAVA

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## TECHNICAL SKILLS

**Languages:** Java, SQL, JavaScript, Python, HTML/CSS

**Frameworks & Libraries:** PyTorch, LangChain, FastAPI, React.js, React Native, Tailwind CSS, Streamlit

**Databases & Vector Stores:** MySQL, MongoDB

**Tools & Platforms:** Git, GitHub, Tableau, VS Code

## PROJECTS

**BrainSegNet – Brain Tumor Segmentation** | PyTorch, Streamlit, FastAPI, Hugging Face, NumPy, Pandas 🌀 **May 2025**

- Achieved a **93% tumor detection accuracy**, **0.87 Dice score**, and **0.17 test loss** on a test set of **531 samples** from the LGG scan dataset by designing and training a custom **U-Net** in **PyTorch** for pixel-wise segmentation of brain tumor regions in MRI scans.
- **Improved generalization** and reduced **overfitting** through extensive data augmentation with **Albumentations**, addressing severe class imbalance.
- Processed **3,900+ MRI slices** across training, validation, and test sets to ensure robust generalization and reproducibility of segmentation results.
- Developed a **FastAPI** backend for MRI tumor segmentation, containerized with **Docker** for future deployment, and integrated a **Streamlit UI** for local, interactive visualization.
- Uploaded the trained model to **Hugging Face Hub** (🌀) for public access and reproducibility.

**ClipQ – YouTube RAG Assistant** | LangChain, Gemini, HuggingFace, FastAPI, Streamlit, Chroma 🌀 **June 2025**

- **Enabled paste-a-link Q&A** over any YouTube video: fetch captions via `youtube-transcript-api`, chunk and embed them, and store vectors in **Chroma**.
- **Delivers answers** by retrieving the **top-k relevant chunks** and prompting **Gemini** to generate responses.
- Exposed the pipeline through clean **FastAPI** endpoints and a minimalist **Streamlit UI** so users can paste a URL, ask questions, and view answers instantly.

**Attendance Tracker App** | React Native (Expo), Notifee 🌀 **April 2025**

- Developed a cross-platform attendance tracker in a 2-person team; owned the **React Native** front-end with screens to mark, edit, and review daily attendance.
- Designed an **intuitive, user-friendly UI** that makes marking and viewing attendance straightforward.
- Programmed lecture-specific reminders with **Notifee**, scheduling **local push notifications** before every class to improve on-time check-ins.
- Deployed the app to Google Play Store for public access (▶️).

## RELEVANT COURSEWORK

- |                                |                    |                     |                               |
|--------------------------------|--------------------|---------------------|-------------------------------|
| • Data Structures & Algorithms | • OOPS             | • Operating Systems | • Computer Networks           |
| • Database Management          | • Machine Learning | • Data Analytics    | • Natural Language Processing |

## EDUCATION

**Sardar Patel Institute of Technology**

*Bachelor of Technology in Computer Engineering – CGPA : 8.39*

Nov 2022 – June 2026

*Andheri West, Mumbai*

**Patkar Varde College**

*HSC - 12th Science – Grade : 90.5%*

June 2020 – May 2022

*Goregaon West, Mumbai*

**Children's Academy**

*SSC - 10th Standard – Grade : 91%*

June 2011 – June 2020

*Malad East, Mumbai*